

REPORT

Student Name: Nguyễn Gia Phúc

Student ID: 524K0011

Class: 24K50201

I. Introduction

- The objective of this lab is to demonstrate the separation of an application into three independent layers: Presentation, Business Logic, and Data Access. By utilizing a 3-Tier Architecture, we improve software maintainability, scalability, and organization, ensuring that each layer maintains a single responsibility and communicates through defined interfaces.

II. Exercises

I successfully implemented the following steps for this exercise:

- Database Setup: Created the exc1 database and the Customers table. I also configured the connection string to establish a link between the application and the SQL Server.
- Layer Development: Developed the core logic by creating the Customer (Model), CustomerRepository (Data Access), and CustomerService (Business Logic) classes.
- UI Design: Designed a Windows Form interface that enables users to input "Name" and "Email," while utilizing a DataGridView to display the retrieved customer list.
- Troubleshooting & Debugging: During development, I identified and resolved several critical issues, including:
 - Entity Framework connection strings errors.
 - Missing DLL build errors.
 - SQL Server conflicts where the table already existed.
- I chose to consolidate the requirements for Exercises 2 and 3. I migrated the Login feature from Lab 2 and refactored it to follow the 3-tier architecture. To optimize development time, I reused and linked the existing SQL database from Lab 2 instead of initializing a new one.

Form1

Name

Email

CustomerID

	Id	Name	Email
	1	John Doe	john.doe@tdtu.c...
	2	Nguyen	Nguyen@tdtu.com
▶	3		
	4	mot	mot@tdtu.com

Developed by Gia Phuc

Form1

Username

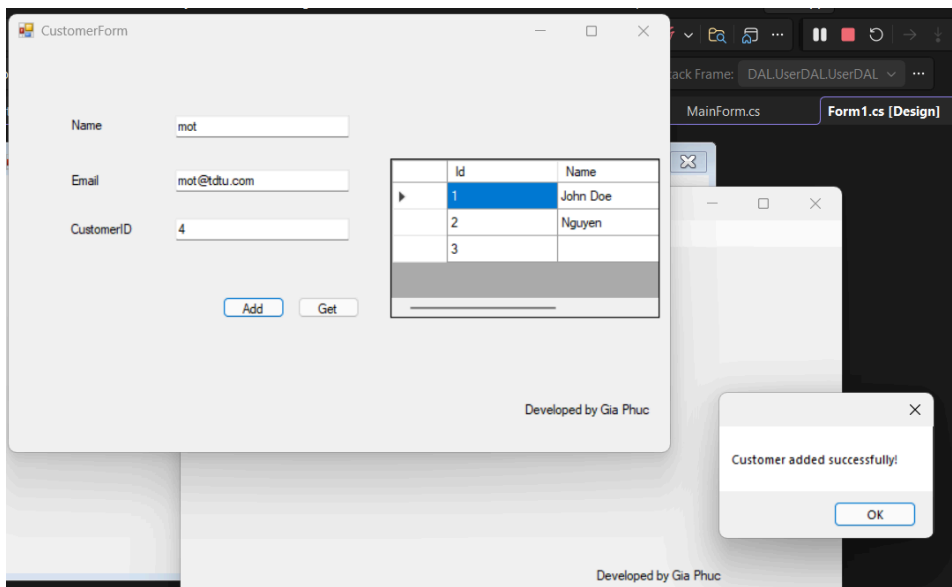
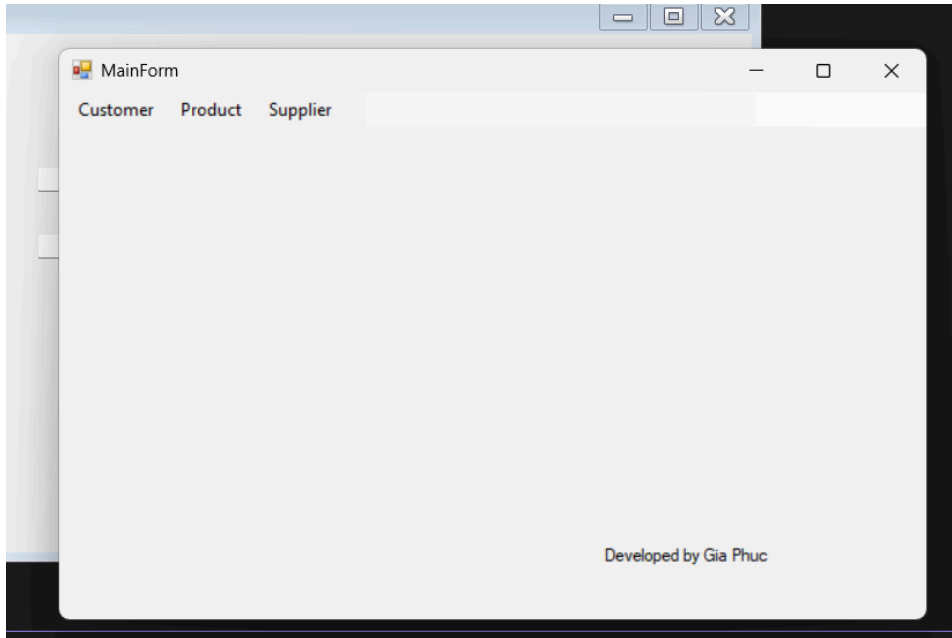
Password

Developed by Gia Phuc

Form1.cs [Design]

Login failed!

OK



CustomerForm

Name

mot

Email

mot@tdtu.com

CustomerID

4

Add

Get

	Id	Name
▶	1	John Doe
	2	Nguyen
	3	
	4	mot

Developed by Gia Phuc